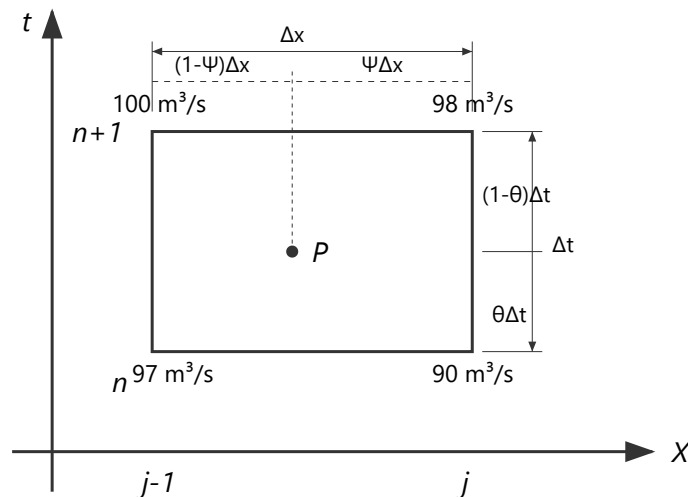


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2080 Chaitra (Regular)

Subject: - Computational Techniques in Civil Engineering (CE 751)

1. Derive the expression for first-order accurate implicit finite difference equation for the kinematic wave model in the linear form. Also, derive the expression for parameters of the momentum equation of the kinematic wave model. [6+2]
2. Determine the values of first order partial derivative of discharge with respect to time and space using four-point implicit method. The discharges at four points in the space-time grid are shown in figure below. Take $\Delta t = 1$ hour, $\Delta x = 500$ m, $\theta = 0.55$, $\Psi = 0.45$. [2+2]



3. Define Method of Characteristics (MOC). Derive the expressions for the application of MOC to unsteady pipe flow problems. [1+7]

4. a). Draw a neat sketch presenting the finite difference grid, showing possible boundary conditions, for 2D steady state seepage analysis under a dam. [2]

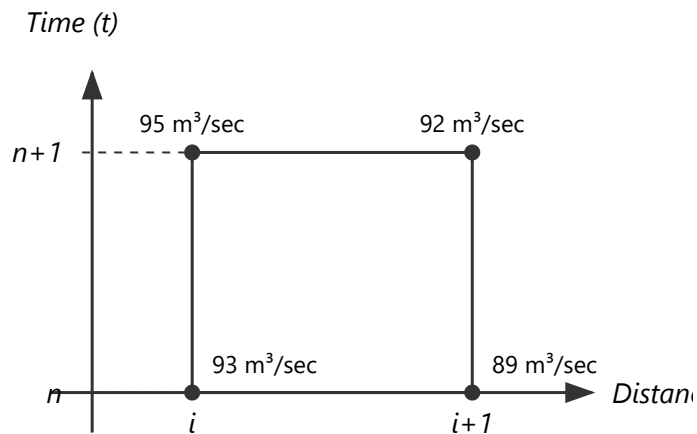
4. b). Table below shows the 2D grid with values of a potential head in m for simulating groundwater seepage. Determine the vertical and horizontal seepage into and out of the highlighted grid. Transmissivity in the x-direction and y-direction are $2700 \text{ m}^2/\text{day}$ and $3200 \text{ m}^2/\text{day}$, respectively, for all grids. Use $\Delta x = 85 \text{ m}$ and $\Delta y = 75 \text{ m}$. [6]

0	0	0	0
2.99	2.92	2.81	2.08
2.97	2.83	2.66	2.51
2.92	2.74	2.53	2.27
0	0	0	0

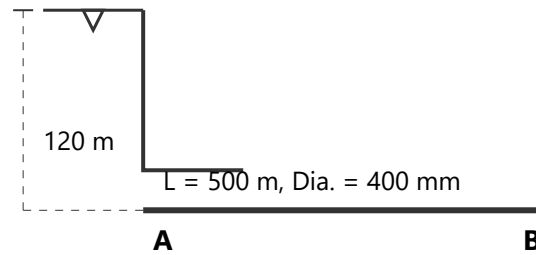
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1. Derive the expression for Leap frog explicit finite equation for dynamic wave modal. [6]
2. The value of flow rate Q at four points in the space time grid are shown in the figure below. Determine the value of first-order derivatives $\frac{\partial Q}{\partial t}$ and $\frac{\partial Q}{\partial x}$ by using four-point implicit method. Given: $dt = 1 \text{ hr}$, $dx = 500 \text{ m}$ and $\theta = 0.47$. [6]



3. A pipe conveys water from a reservoir as shown in the figure below. Take $f = 0.02$, $C = 1200 \text{ m/s}$. The HGL at the reservoir is given as $H_{PA} = 120 + 3 \sin(\pi t)$. The discharge at the downstream end is zero at all times. By using only one reach, compute discharge from A and elevation of hydraulic grad line at B at 3 sec using a discretized equation of the method of characteristics in the form of HGL and discharge. [8]



4. Derive finite difference equation to evaluate river stage-water table interaction for one dimensional flow along a river and express in terms of tridiagonal coefficient matrix. [8]

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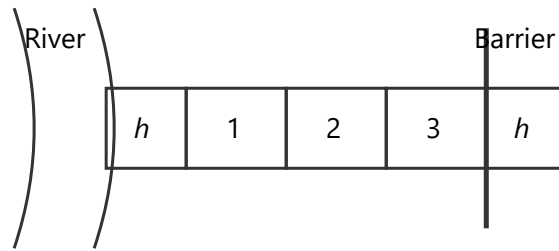
1. Write down the governing equations used for analyzing the moment of fluid. [2+3]
Discuss forward, backward and central differencing with expressions.

2. Following are data pertaining to a rectangular channel: [6]
- Width of channel = 200 ft
 - Length of Channel = 15000 ft
 - Bed Slope, $S_0 = 1\%$
 - Manning's $n = 0.035$

At the $t = 0$, there is a uniform flow of **2000 cfs** along the channel. The discharge value at the upstream boundary from inflow hydrograph at time $t = 3 \text{ min}$ is obtained as **2250 cfs**. Determine the discharge at a distance of 3000 ft downstream along the channel. Use the linear kinematics wave modal. Take $\Delta x = 3000\text{ft}$ and $\Delta t = 3 \text{ min}$. There is no lateral inflow ($q = 0$).

3. What do you understand by Method of characteristics? Derive the finite difference form of characteristic equations for unsteady pipe flow in terms of head and discharge. [2+6]

4. a). A schematic for simulating river stage water table fluctuation is shown in figure. The following data are given for the simulation of homogenous and isotropic aquifer, river stage (h_L) = 365m, length of aquifer = 1200 m, Δt = 1 day, Δx = 400 m, transmissivity of aquifer = $600 \text{ m}^2/\text{day}$, storage coefficient = 0.15. The initial value of water table at 3 grids as 349.13, 347.97, 339.36 respectively. Calculate water table elevation in each grid. [5]



4. b). Explain the concept of finite difference method. What are explicit and implicit schemes in finite difference method? [2+1+1]